Cover letter — IEEE Access submission

To: The Editors, *IEEE Access* **Article:** *Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is — in Simulation and on $30 of Silicon* **Author:** Mulham Fetna, Department of Mechatronics Engineering, University of Aleppo, Aleppo, Syria · ORCID 0009-0006-4432-798X · contact@mulhamfetna.com

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Dear Editors,

I am pleased to submit the above article for consideration in *IEEE Access*.

What the article asks and answers. It asks whether ambient sub-7 GHz WiFi, received on a moving vehicle and processed with commodity channel state information (CSI), can replace radar or LiDAR as a front-end for simultaneous localization and mapping. The answer is a boundary rather than a yes or no, and that boundary is the contribution: WiFi localizes to centimetre level — at parity with a mid-range LiDAR for roughly one percent of its cost — while WiFi *mapping* hits a hard ceiling. The article locates that ceiling precisely: approximately 89 % of realistic-CSI detections correspond to no real propagation path, accompanied by a several-metre range bias. It is a front-end and geometry limit, not a path-discrimination limit, not a resolution limit, and not repairable by a learned filter.

Why it should interest a broad readership. Running the same substrate against real LiDAR (KITTI) and a simulated 77 GHz FMCW radar shows the ceiling is neither carrier-specific nor WiFi-specific: the carrier does not matter, geometry does — a monostatic, on-vehicle transmitter reduces the phantom rate from 18.2 % to 0.1 % — and wider bandwidth makes phantoms worse. The article also reports a cost analysis (the WiFi package is 84–600× cheaper than the simulated LiDAR class), a symmetric fusion study yielding a design rule (fusion reinforces at sensor parity and degrades the stronger sensor eightfold when mismatched), a robustness characterization, and a hardware demonstration on two $30 ESP32-S3 boards.

Reproducibility. All code, configurations, result data, and the ESP32 firmware are released openly under AGPL-3.0 at https://github.com/mulhamfetna/wifi-radar-slam and archived with persistent DOIs. Every reported number is regenerable from that artifact.

Disclosure of similarity to my own prior work. This article consolidates and substantially extends earlier **unpublished** manuscripts of my own — a physics-based feasibility study and a WiFi-versus-LiDAR comparison. Those drafts are cited in the article and are publicly readable in the open-source repository above, so an automated similarity check may match text against them. To be explicit: they are my own unpublished work, not prior publications, and this submission is a consolidated and considerably extended treatment that adds the radar comparison, the robustness experiments, the hardware validation, and a corrected analysis. I note also that a numerical result reported in the earlier feasibility draft did not reproduce on re-verification; the corrected value (absolute trajectory error 0.098 ± 0.028 m) is the one used throughout this article.

Originality and exclusivity. The work is original, is not under consideration for peer review at any other journal, and has not been published elsewhere.

Thank you for considering this submission. I would be glad to provide any further information.

Sincerely, **Mulham Fetna** ORCID 0009-0006-4432-798X · contact@mulhamfetna.com